SA 220 AQUACULTURE

REPRODUCTIVE FAILURE OF SOME FISH SPECIES IN CAPTIVITY

SOME DEFINATIONS

Milt- The semen of a male fish (loosely sperm)

Oocyte - A cell in an ovary which may undergo meiotic division to form an ovum.

Gametogenesis- This is the production of sperm (spermatogenesis) and eggs (oogenesis), that occurs through meiosis.

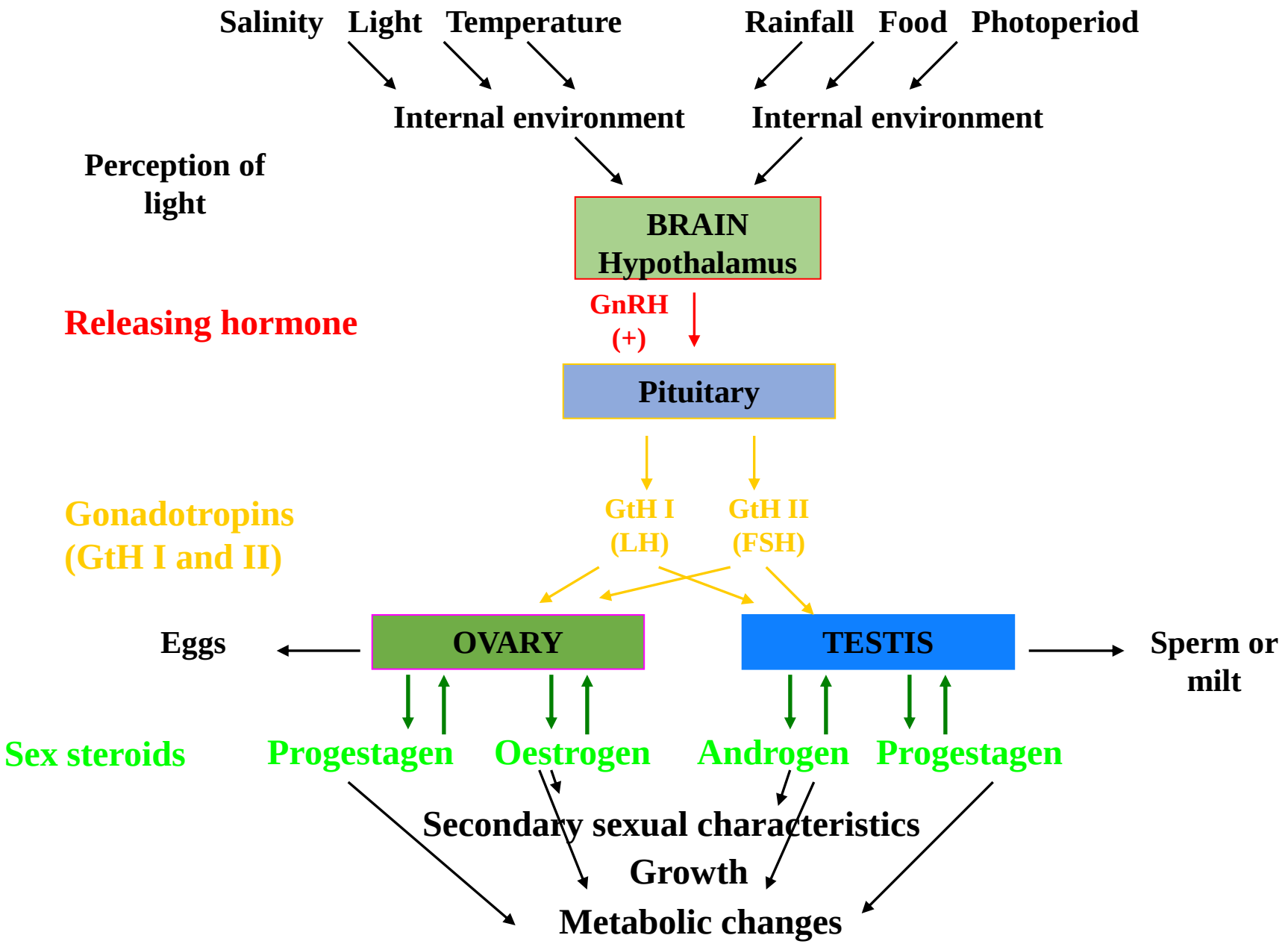
Spermiation- This is the process by which mature spermatids are released from the supporting somatic Sertoli cells.

Ovulation- This is the releasing of mature eggs

Environmental cues in fish reproduction

- Most fish reproduce at specific times of the year.
- The factors the induce reproduction include; The length of the day and temperature.
- Further, the reproduction behavior and sexual maturation need to be In addition, sexual maturation and reproductive behaviour needs to be harmonized between the sexes.
- This can involve visual and chemical signals or communication.

Control of Reproduction



Fish Puberty

- Fish reaches puberty when the Brain-pituitary-gonadal (BPG) axis is fully operatating or functional.
- In most fish species, the puberty starts when the germ cells maturation and functional differentiation of somatic or cells of the gonads which results in spermiation and ovulation.

AQUACULTURE AND FISH PUBERTY

In quaculture, understanding puberty of your fish species is critical because;

1. Early puberty is a common and big problem in a number of fish species.

Why?

AQUACULTURE AND FISH PUBERTY

ANSWER: Early puberty has negative effects on growth performance and other desirable features

AQUACULTURE AND FISH PUBERTY

2. Late puberty, or failure to reach puberty in captivity, can be a problem with broodstock in some species

- Onset of puberty controlled by activation of BPG axis

REPRODUCTIVE FAIL

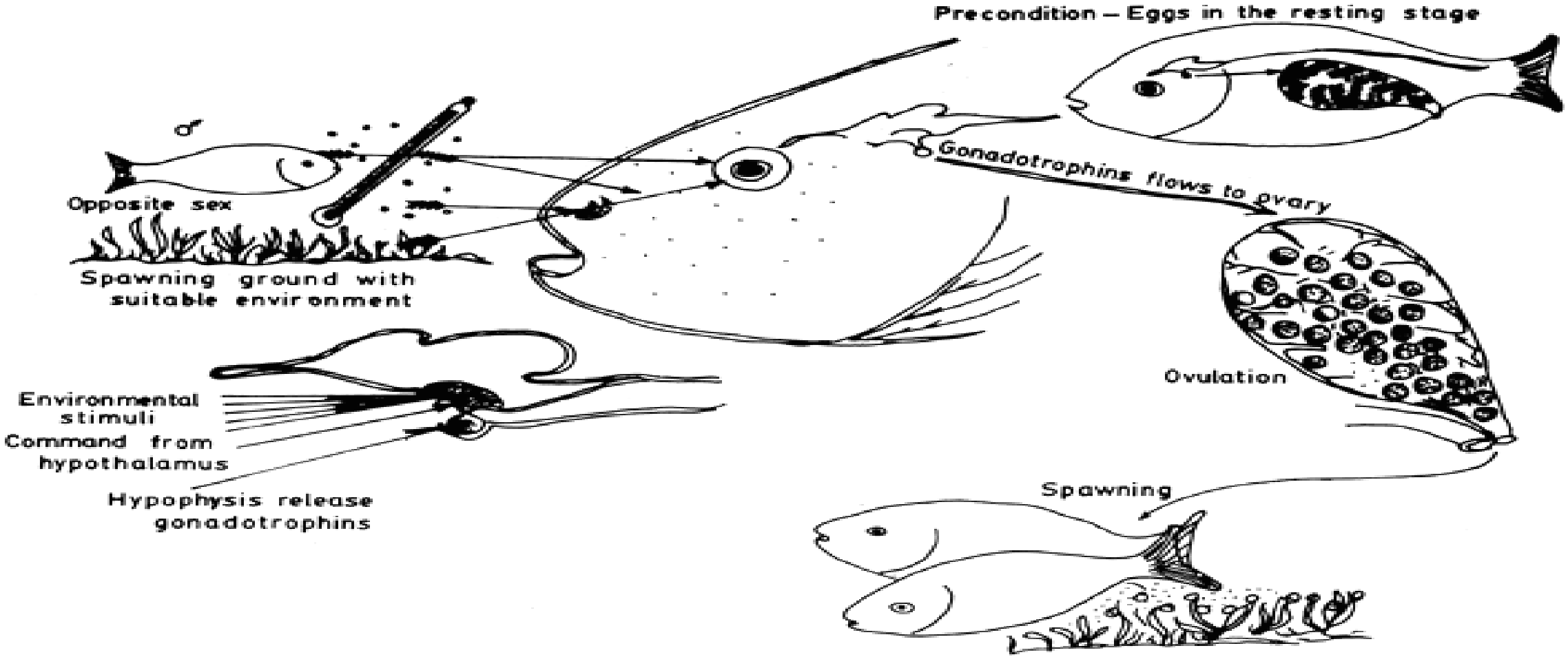
- Some fish reared in captivity exhibit some form of reproductive dysfunction
- This is failure of fish to undergo Final Oocyte Maturation (FOM), ovulation, spermiation and spawning.
- **WHY?**

REPRODUCTIVE FAIL

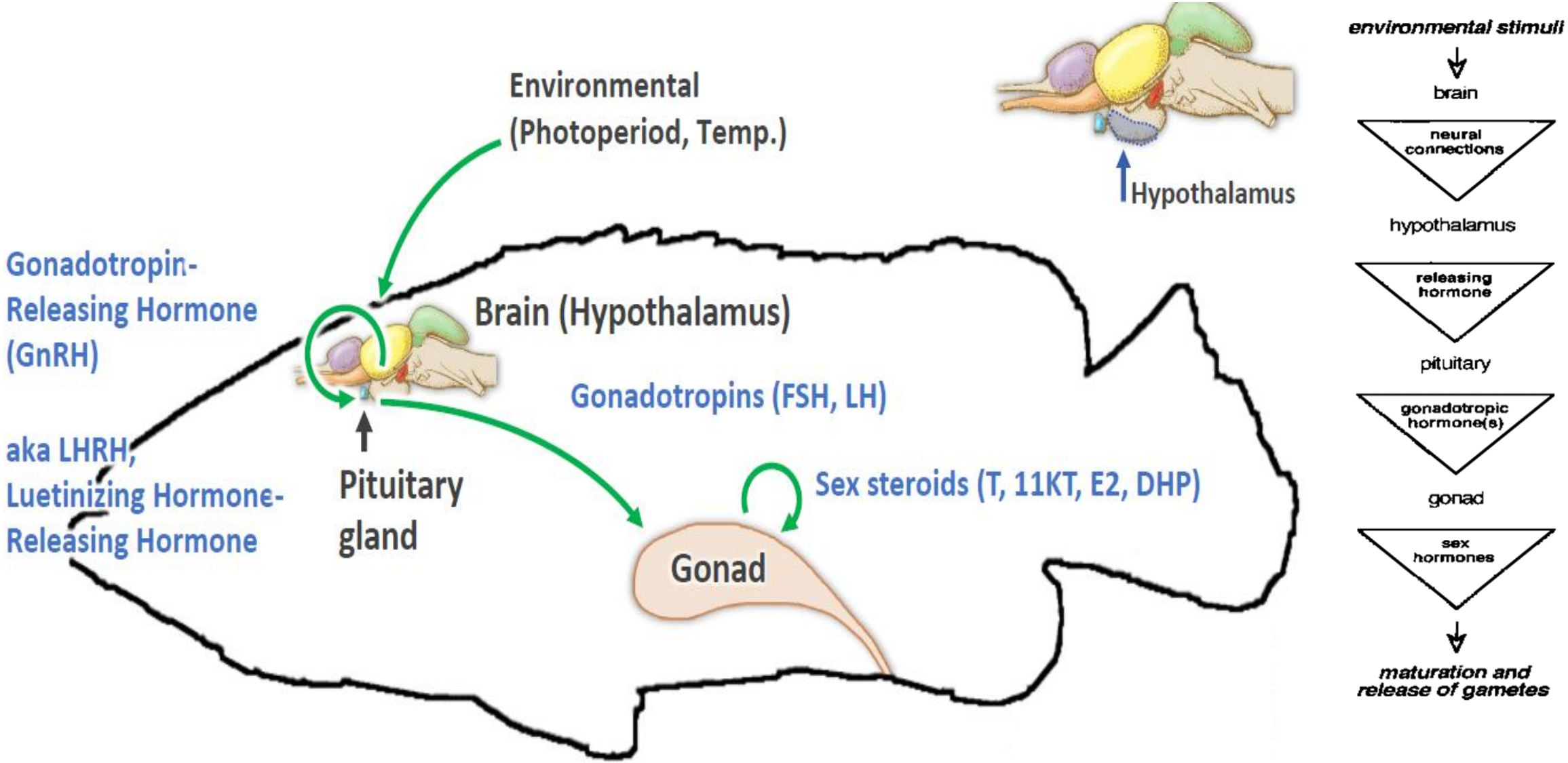
Selected answers:

- Differences in the environmental conditions (water salinity, temperature, depth and substrate)
- These dysfunctions probably result from the combination of captivity-induced stress (Sumpter et al., 1994)
- lack of the appropriate natural spawning environment

NORMAL SITUATION



NORMAL SITUATION



Gonadotropins (GtH)

The pituitary gland produces GtH and these are about 200 Amino Acid.

Example:

GtH I = FSH : Egg growth

GtH II = LH : Final maturation & Ovulation

Act on steroidogenic cells:

- in male : Sertoli cells
- in female : Follicular cells (Theca and granulosa)

Sex steroids

Oestrogen: Oestradiol (E2) in female fish

Androgens: Testosterone (T) in male and Females

Testosterone (T) in males and females

Tak note: The Testosterone in females will be converted to Estradiol by an enzyme Aromatase

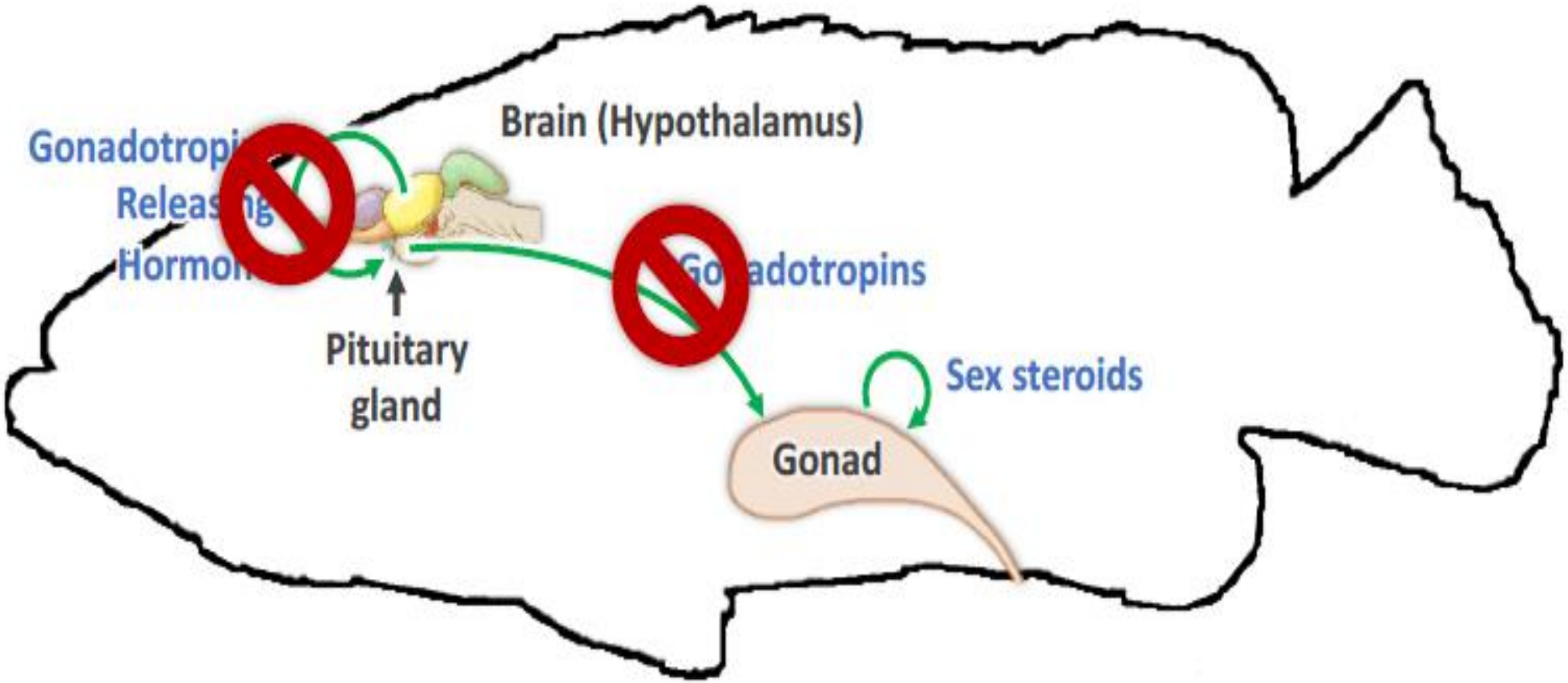
11-ketotestosterone (11-KT) in males

Progestagens

17, 20β-dihydroxyprogesterone (DHP)

17α,20β,21-trihydroxy-4-pregnen-3-one (20 β-S)

ABNORMAL SITUATION (CAPTIVITY)



THREE DYFUNCTIONS IN FEMALE FISH

1. Failure completely to undergo vitellogenesis and spermatogenesis when in captivity

2. Absence of spawning at the end of the reproductive cycle

3. Absence of final oocyte maturation

TWO DYFUNCTIONS IN MALE FISH

1. Reductions in milt production

2. Reductions in milt quality







DYFUNCTIONAL - EXAMPLES

SPECIES	SPONTANEOUS SPAWNING	SPERMATOGENESIS	SPERMIATION	VITELLOGENESIS	OM AND OVULATION
African catfish	NO	YES	NO	YES	NO
Tilapia	YES	YES	YES	YES	YES
Carp	NO	YES	YES	YES	NO
Cod	YES	YES	YES	YES	YES
Sole	NO	NO	YES	YES	NO
Seabass	YES	YES	NO	YES	YES
Salmon	NO	YES	YES	YES	YES
Turbot	NO	YES	YES	YES	YES
[redacted]	STRIPPING	[redacted]	SACRIFICE	[redacted]	HORMONAL INDUCTION

CORRECTING DYFUNCTIONAL

1. Manipulations of various environmental parameters, such as temperature, photoperiod, salinity, tank volume and depth, substrate vegetation (Munro et al., 1990)

- For instance, a sadden increase in water level in a pond with catfish may mimic rainfall and cause the species to go into reproduction mode.

CORRECTING DYFUNCTIONAL



CORRECTING DYFUNCTIONAL

2. hormonal treatments are the only means of controlling reproduction reliably

- DOCA (Deoxycorticosteroid Acetate), 2.5-5 mg per 100 g of female
- HCG (Human Chorionic Gonadotropin), 25 I.U. (international units) per 100 g of female
- Ovaprim or Aquaspawn (1.5ml/kg of fish and half the dose for males).
- pituitary gland material of fish

CORRECTING DYFUNCTIONAL

The pituitary of fish produces two gonadotropins; LH and FSH

- FSH is concerned with follicle development and vitellogenesis
- LH is concerned with final oocyte maturation, ovulation and spawning

DRAWBACKS OF HORMONES

- Variable in potency (efficacy or strength) and has short storage life
- Species-specificity has been shown for fish gonadotropin
- Poor reproductive performance
- High cost of fish pituitaries
- Unavailability of good quality pituitary glands in sufficient quantity

WHY CORRECTING DYFUNCTIONAL ?

- Induction of ovulation and spermiation
- Acceleration of spawning date
- Synchronization of spawning dates between related species
- Production of fry outside the normal spawning

END

THANK YOU FOR YOUR ATTENTION